**Department of Computer Science and Engineering**
**Academic Year 2023 – 2024 (Odd Semester)**

**Degree, Semester & Branch: I Semester B.E. CSE-'B'**
**Course Code & Title: GE3151 Problem solving and python programming**
**Name of the Faculty member (s): Mrs.S.Manjula**

**Innovative Practice Description**

**Unit / Topic: Unit IV / Introduction to List, Lists: list operations, list slices, Dictionaries: operations**

**Course Outcome:** CO 4

**Topic Learning Outcome:** TLO 11

**Activity Chosen:** Video Lecture and Mind Map

**Justification:**

- Lists are important because many programs like linear and binary search work with lists. Furthermore, proper understanding will help you learn and understand program logic easily. To become familiar with the topic, a video lecture was created and posted on the department's YouTube channel one week before the actual lecture, and the students were instructed to watch the video (https://youtu.be/5gEQerenQAw). It helps students by making concepts more interesting and motivating them to learn more about a specific topic.
- Time Allotted for the Activity: 40 minutes
- A mind map is a visual representation of ideas and concepts. It's a visual thinking tool that aids in the organization of data. It jots down ideas quickly. As a result, this activity will assist students in identifying the theoretical concepts in the image as well as aid in easy recall for their exams.
- **Time Allotted for the Activity:** 15 minutes

**Detail Implementation of Video Lecture Activity:**

- The students were informed about the event, and a video lecture was posted 1 week in advance so that they could contribute enough time to learning the subject.
- The instructor explained the importance of learning by doing to the students and informed them that it is an educational approach to program-based learning.
- A student was chosen at random to explain the concept to their peers using any of the teaching aids.
- Rupalakshmi A of the class explained the list accessing and updating.
- Selvakiruthiga C explained how various operations on the list were performed with examples.
- Finally, the instructor consolidated the information that was discussed in this activity.
- This assisted the students in learning about themselves and answering questions about the topic with ease.

**Detail Implementation of Mind Map:**

- Instructor explained the particular concepts/topic in classroom within 35 minutes.
- Based on the discussion and the teacher asked the students to draw a mind map related to the topic within 15 minutes.
- Each student created a mind map on Dictionaries: operation and methods.
- Students represent the concept taught in the class by visual representation.
- Instructor collected the sheets from the students.

**CO – PO / PSO mapping:**

| CO | PO1 | PO2 | PO3 | PSO1 |
|---|---|---|---|---|
| CO 4 | 2 | 1 | 1 | 1 |

**(1 – Low  2 – Moderate  3 – High)**

**PO / PSO mapped:**

| Innovative practice | PO1 | PO2 | PO3 | PSO1 |
|---|---|---|---|---|
| | 2 | 1 | 1 | 1 |
| **Justification for correlation** | Students can able to use the data type dictionary and List for providing solution to the problems by using basics mathematics | Data types can be identified, formulated to solve the simple and complex engineering problems | Students can be able to provide solution to the real time decision making problems by using necessary compound data types. | Students can be able to apply compound data types in python program to provide the solution for information and communication technology related problems. |

- **Images / Screenshot of the practice:**

**Glimpses of Video Lecture Activity**


*Figure 1: Glimpses of List Accessing Explanation by Rupalakshmi A*

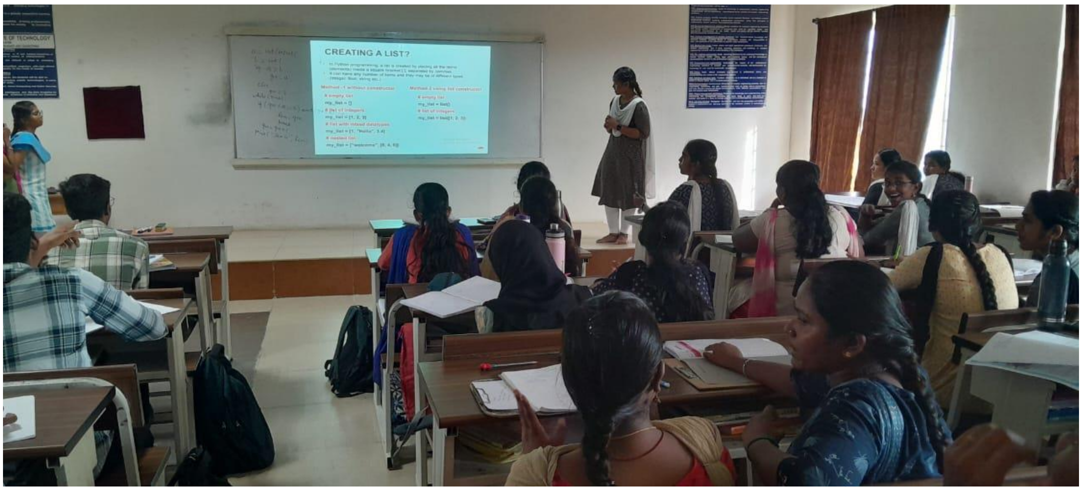


*Figure 2: Glimpses of List operation Explanation by Selvakiruthiga C*

**Glimpses of Mind Map Activity**

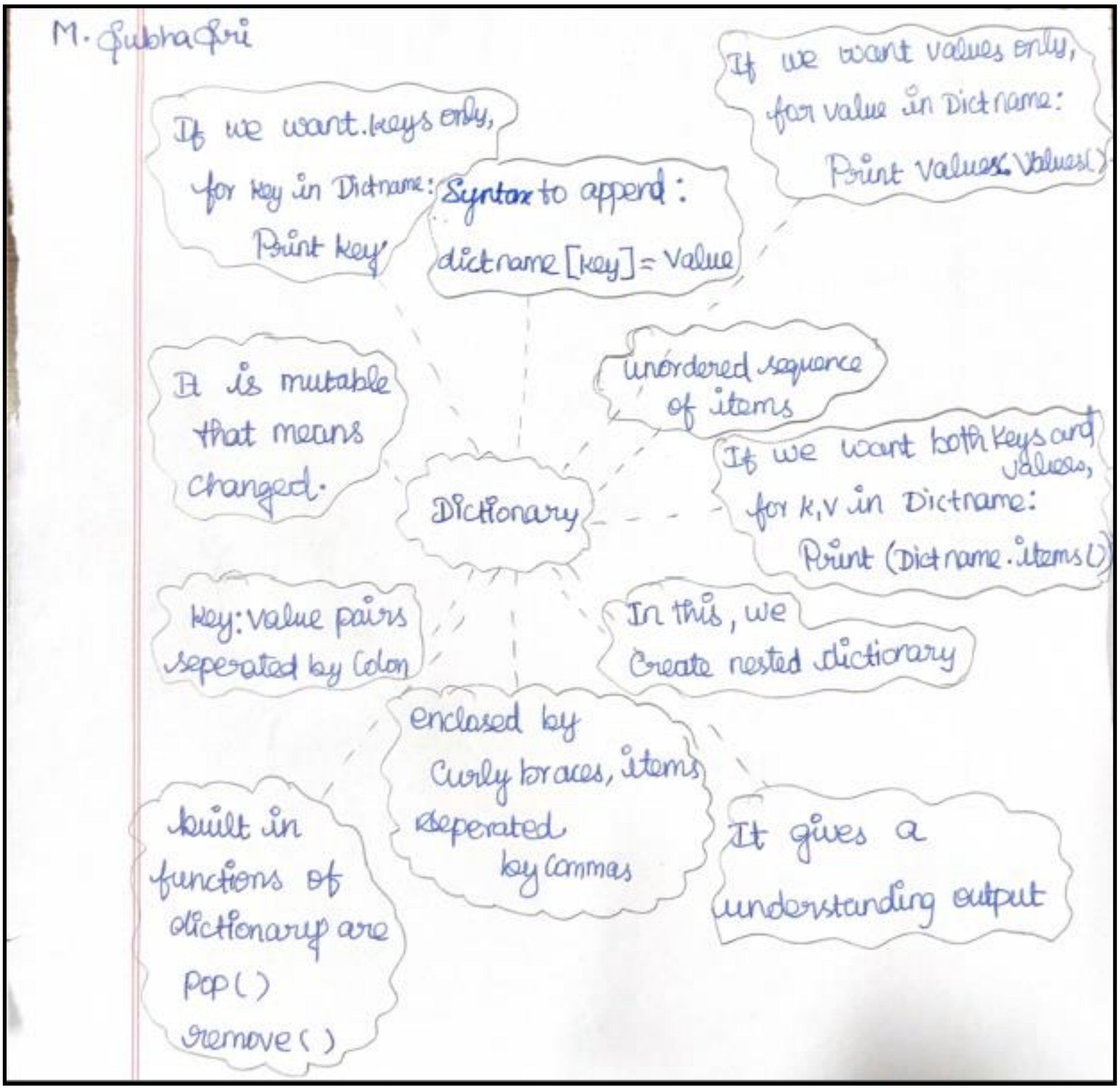


*Figure:3 Mind Map by Subhasri M*

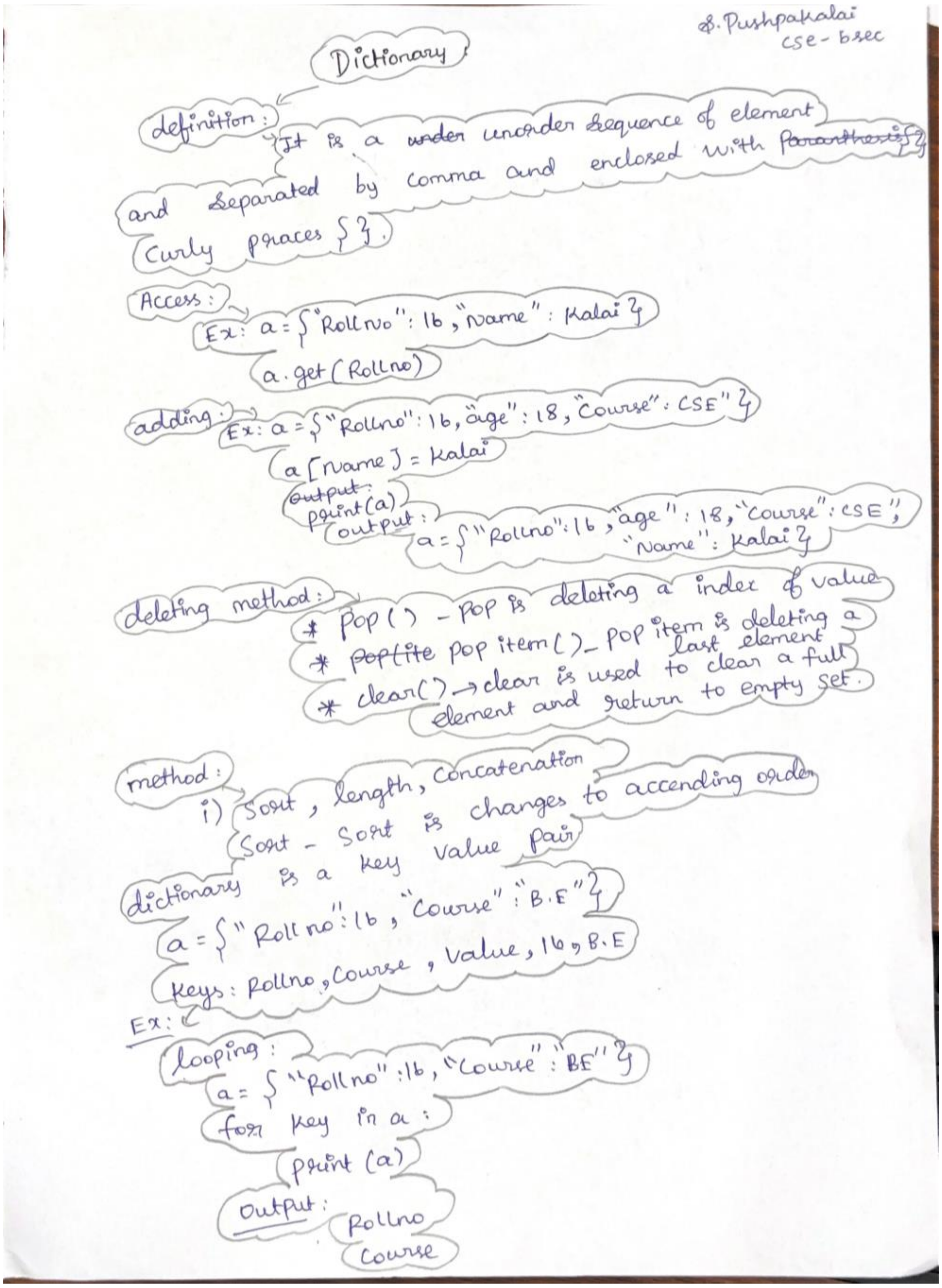


*Figure :4 Mind Map by Pushpakali S*

- **Reflective Critique:**
    - ❖ ***Feedback of practice from students and other stakeholders:***

        - Students indicated to the teacher that the exercise encourages students to study independently, clear up any questions, and share ideas with their classmates. *(Video Lecture)*
        - Students expressed that they understood the concept very well and the pictorial representation made them to understand the topic clearly. *(Mind Map)*
        - This helped the students in recalling the topic taught on that specific day, generating new ideas about the topic. *(Mind Map)*

❖ ***Benefit of the practice:***

- Students spent their time in self-learning. *(Video Lecture)*
- From this activity, the students can get more clarity in the particular topic. *(Mind Map)*
- Through this activity, the students are able to remember and describe the various dictionary operations and it methods clearly. *(Mind Map)*

❖ ***Challenges faced in implementation:***

- Effectively motivate students who are not participating in the activity by addressing the benefits of self-learning. *(Video Lecture)*
- Some of the students feel difficult to depict the concepts pictorially. *(Mind Map)*
- Some of them need some more clarification on the dictionary operations. *(Mind Map)*
- Additional examples given to make them to understand the concept easily. *(Mind Map)*

**References:**

https://www.ritrjpm.ac.in/images/computer-science/Mind%20Map.pdf
https://www.ritrjpm.ac.in/images/computer-science/43.CS6703_MindMap.pdf
https://www.ritrjpm.ac.in/images/computer-science/5_CS8591_Mindmap.pdf
https://www.lucidchart.com/pages/how-to-make-a-mind-map

https://www.ritrjpm.ac.in/images/computer-science/2021-2022/BV_Mindmap.pdf